

University of Wah

Department of Computer Science



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Course:

Data Structure and Algorithm Lab

Reg No:

UW-24-CS-BS-101

Section:

BS-CS-3rd-B

Lab 13

Task

Code

```
#include <iostream>
#include <vector>
using namespace std;

void heapifyMax(vector<int>& heap, int n, int i) {
    int largest = i;
    int left = 2*i + 1;
    int right = 2*i + 2;

    if (left < n && heap[left] > heap[largest])
        largest = left;

    if (right < n && heap[right] > heap[largest])
        largest = right;

    if (largest != i) {
        swap(heap[i], heap[largest]);
        heapifyMax(heap, n, largest);
    }
}

void buildMaxHeap(vector<int>& heap) {
    for (int i = heap.size()/2 - 1; i >= 0; i--)
        heapifyMax(heap, heap.size(), i);
}
```

```
void executeTasksMaxHeap(vector<int> heap) {
    buildMaxHeap(heap);

    cout << "Task Execution Order : ";
    for (int i = heap.size() - 1; i >= 0; i--) {
        cout << heap[0] << " ";
        swap(heap[0], heap[i]);
        heapifyMax(heap, i, 0);
    }
    cout << endl;
}

int main() {
    vector<int> tasks = {5, 3, 10, 1, 4};
    executeTasksMaxHeap(tasks);
    return 0;
}
```

For Max

```
Task Execution Order : 10 5 4 3 1  
[1] + Done
```

Modify Code for Min

```
void heapifyMin(vector<int>& heap, int n, int i) {  
    int smallest = i;  
    int left = 2*i + 1;  
    int right = 2*i + 2;  
  
    if (left < n && heap[left] < heap[smallest])  
        smallest = left;  
  
    if (right < n && heap[right] < heap[smallest])  
        smallest = right;  
  
    if (smallest != i) {  
        swap(heap[i], heap[smallest]);  
        heapifyMin(heap, n, smallest);  
    }  
}
```

For Min

```
Task Execution Order : 1 3 4 5 10  
[1] + Done
```

Dry Run

~ Dry Run ~

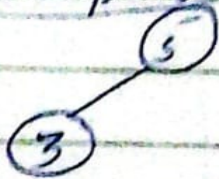
tasks = [5, 3, 10, 1, 4]

Min r

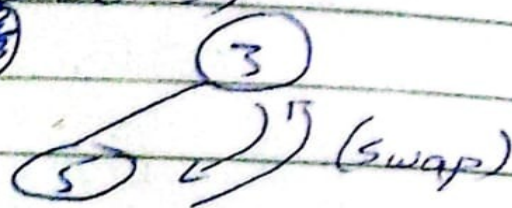
Step 1 (Insert 5)

(5)

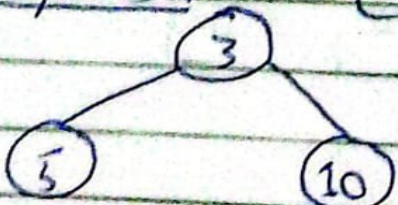
Step 2:- (Insert 3)



$5 > 3$ ✗

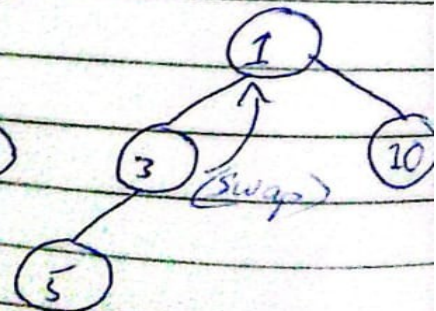
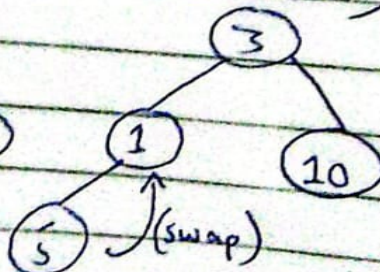
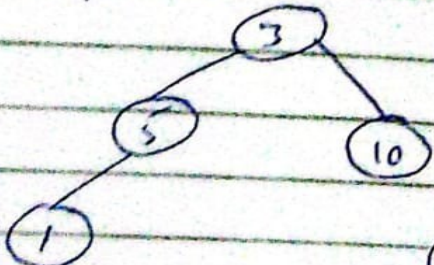


Step 3 (Insert 10)



$3 < 5 < 10$

Step 4 (Insert 1)



DATE: _____

DAY: _____

Step 5

(Insert 4)

